LESSON 1 • TERM 4 • LEVEL 1

Half and Quarter

Equal parts: halves and quarters

TODAY WE WILL

Fold shapes into equal parts.



How our lessons work

0–4 MIN

1. Listen first, then try. A guess is a good start.
2. Answer on three: say it quietly, show fingers or a thumbs up.
3. Tick your own work on the answer slide.
4. Still stuck? Tick the box on your look-back page and show your grown-up.

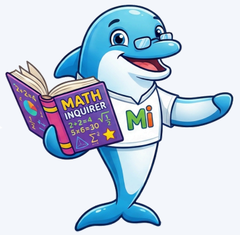
Warm-up: share 8

4–10 MIN

Share 8 dots between 2 plates. Both plates must match.



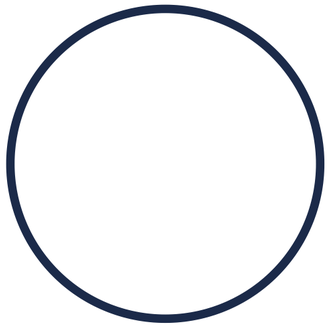
How many dots go on each plate?



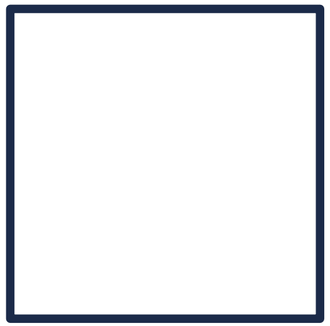
Look at both plates!

Name each shape

4–10 MIN



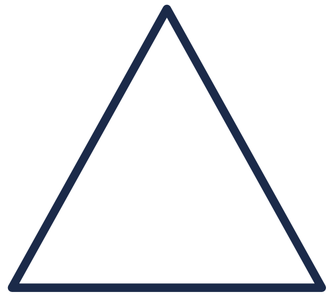
A



B



C



D

Say the name of each shape. How many sides has each one?

Warm-up: share 7

4–10 MIN

Share 7 dots between 2 plates.



Can both plates match?

Sneak-peek puzzles

10–14 MIN

PUZZLE 1: anyone can try

Fold a sheet of paper so the edges meet. Open it. How many parts has the fold made?

PUZZLE 2: think

Fold a sheet of paper so the edges meet. Fold it the same way again. Open it. How many equal parts are there? I know because ___.

Explore: fold a circle

14–26 MIN

1. Take one paper circle from your kit.
2. Fold it so the edges meet.
3. Open it. Trace the fold with a pencil.
4. Lay one part on top of the other. Do they match?

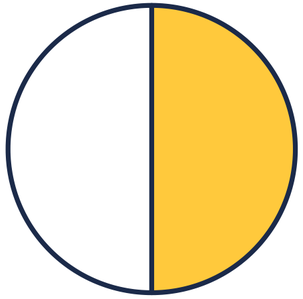
Fold only. Nobody cuts.

Explore: fold a circle

14–26 MIN

1. Take one paper circle from your kit.
2. Fold it so the edges meet.
3. Open it. Trace the fold with a pencil.
4. Lay one part on top of the other. Do they match?

Fold only. Nobody cuts.



2 EQUAL PARTS

Each part is a half.

2 halves make 1 whole.

Explore: fold again

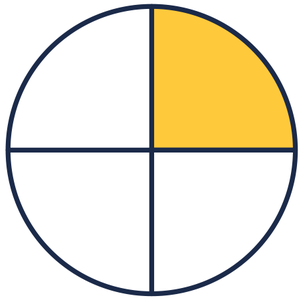
14–26 MIN

1. Fold your circle in half again.
2. Open it. Trace the new fold. How many parts now?
3. Lay the parts on top of each other. Do they all match?
4. Now take a long rectangle. Fold it in half, then in half again. Open it.

Explore: fold again

14–26 MIN

1. Fold your circle in half again.
2. Open it. Trace the new fold. How many parts now?
3. Lay the parts on top of each other. Do they all match?
4. Now take a long rectangle. Fold it in half, then in half again. Open it.



4 EQUAL PARTS

Each part is a quarter.

4 quarters make 1 whole.

2 quarters make 1 half.

Explore: a crooked fold

14–26 MIN

1. Take your second circle.
2. Fold it so the edges do not meet.
3. Open it. Trace the fold.
4. Lay one part on the other. Do they match?

Two parts, but are they halves?

Explore: a crooked fold

14–26 MIN

1. Take your second circle.
2. Fold it so the edges do not meet.
3. Open it. Trace the fold.
4. Lay one part on the other. Do they match?

Two parts, but are they halves?

The parts do not match. They are not equal, so they are not halves.

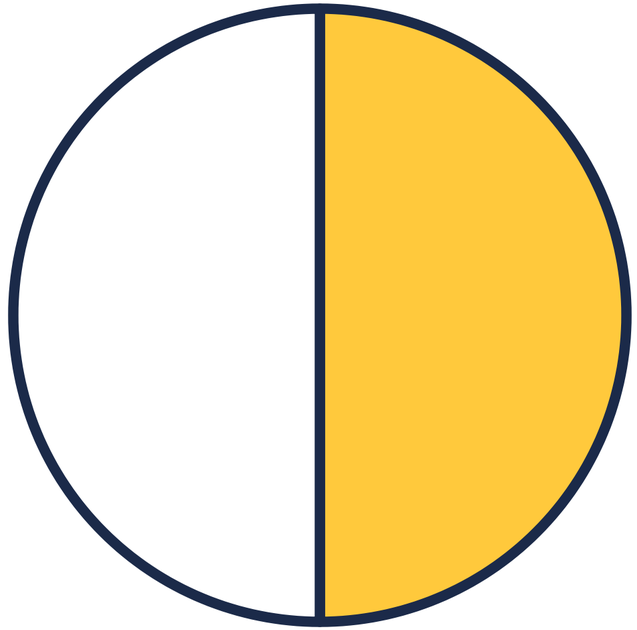
A half

26–36 MIN

A HALF

One of 2 equal parts.

2 halves make 1 whole.



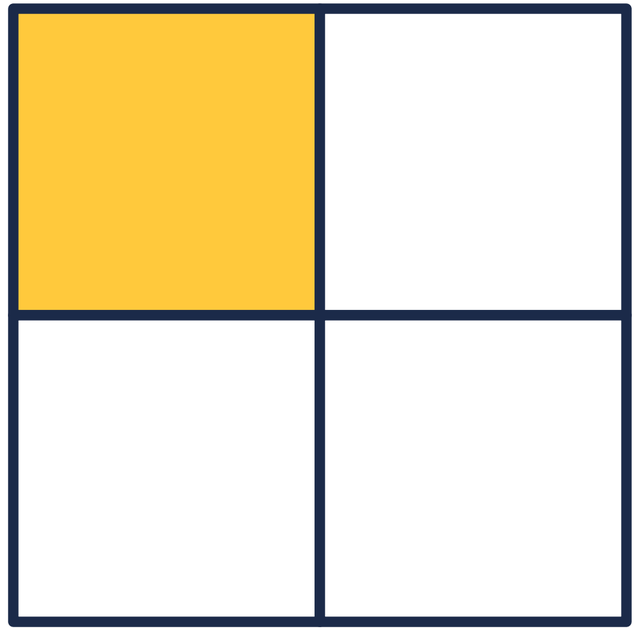
A quarter

26–36 MIN

A QUARTER

One of 4 equal parts.

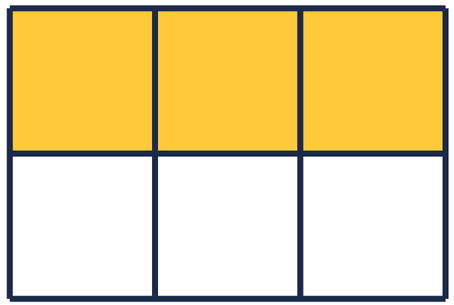
4 quarters make 1 whole.



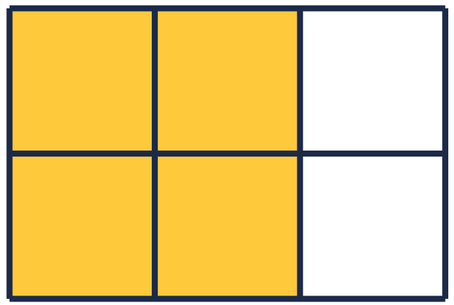
Is half coloured?

26–36 MIN

Count the coloured squares and the white squares.



Picture 1



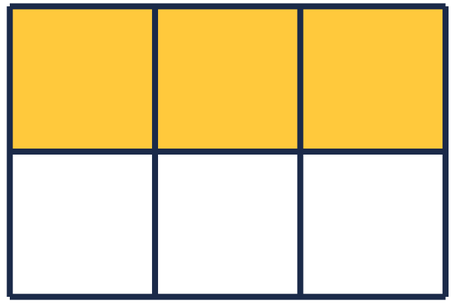
Picture 2

In which picture is half coloured?

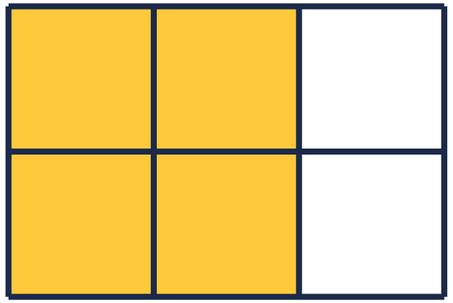
Is half coloured?

26–36 MIN

Count the coloured squares and the white squares.



Picture 1



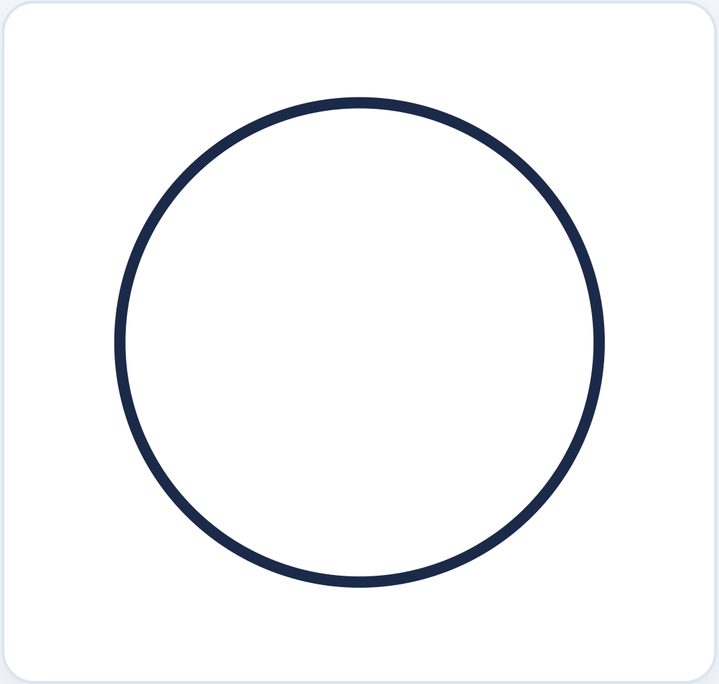
Picture 2

In which picture is half coloured?

Picture 1: 3 coloured and 3 white match, so half is coloured. Picture 2: 4 coloured and 2 white do not match.

Half circle and quarter circle

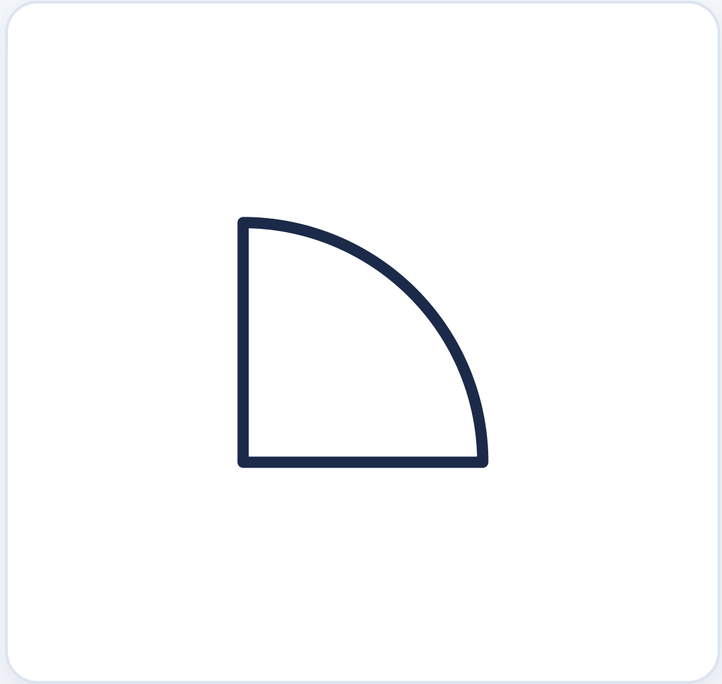
26–36 MIN



whole circle



half circle



quarter circle

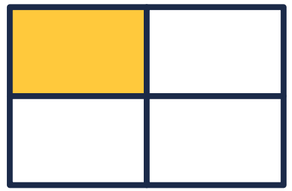
A half circle is half of a circle. A quarter circle is a quarter of a circle.

Your turn

26–36 MIN

ITEM 1

Is the coloured part a half or a quarter?



ITEM 2

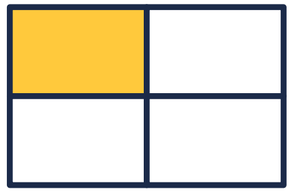
Which is bigger, a half or a quarter of the same circle?

Your turn

26–36 MIN

ITEM 1

Is the coloured part a half or a quarter?



ITEM 2

Which is bigger, a half or a quarter of the same circle?

Item 1: a quarter. The rectangle has 4 equal parts. Item 2: a half. 2 parts are bigger than 4 parts of the same circle.

36–40 MIN

Turn like a clock

Stand up. Face the screen.

A half turn. A quarter turn.

Sip some water.

Workbook page 1: Start here

40–70 MIN

Questions 1 to 3. Take your time.

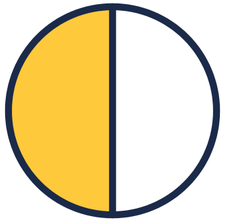
S1 Look at the three shapes. Is half of each shape coloured? Circle Yes or No for each shape.



A



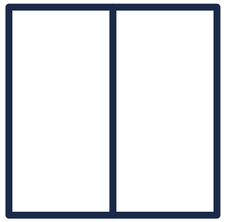
B



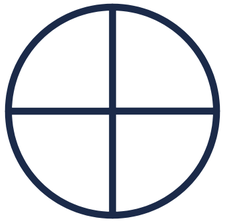
C

S2 Look at shape B in question 1. Fill in the blanks.

S3 Colour the shapes.



square



circle

Check page 1

40–70 MIN

S1	A yes; B no; C yes
S2	4 equal parts; quarter
S3	any one part of the square; any one part of the circle

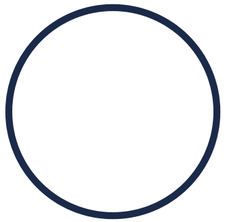
Workbook page 2: Start here

40–70 MIN

Questions 4 to 6. Take your time.

S4 Fill in the blanks.

S5 Look at the four shapes. Write the number of the shape.



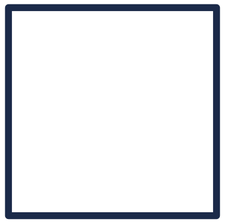
1



2

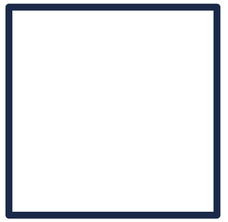


3

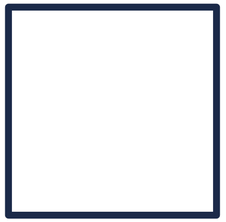


4

S6 Use your ruler. Draw straight lines on the squares.



A



B

Check page 2

40–70 MIN

S4	2; 4
S5	2; 3
S6	square A: 1 line through the middle; square B: 2 lines through the middle that cross

40–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Cut a square in different ways
Go further (page 4)	Same Half, Different Shape
Investigation (page 5)	Fold twice
Check	Finish and check your own work

Everyone may choose. Not finished? Pages 3 to 5 are for home too.

Talk and explain

70–76 MIN

A CHILD SAYS

Any 2 parts of a shape are halves.

Is the child right? Test it with your circle.

I think ... because ...

Back to the sneak-peek

70–76 MIN

Puzzle 1: 2 parts. Puzzle 2: 4 equal parts. The second fold cuts each of the 2 parts into 2.

Exit ticket

76–82 MIN

On your own. No looking back.

E1 A circle is cut into 4 equal parts. Is each part a half or a quarter? Circle one.

E2 Fill in the blank.

☐ halves make 1 whole.

E3 A square is cut into 2 parts that are not the same size. Are they halves? Circle Yes or No.

Check the exit ticket

76–82 MIN

E1	quarter
E2	2
E3	No

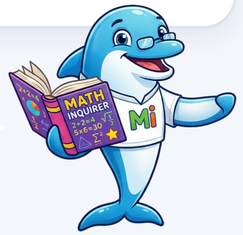
Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ folding the paper ☐ knowing if parts are equal ☐ half or quarter ☐ something else
What helped me	☐ my paper shapes ☐ laying one part on the other ☐ checking my answer ☐ I was not stuck
Next time I will try	☐ folding carefully ☐ checking the parts match ☐ checking my answer ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

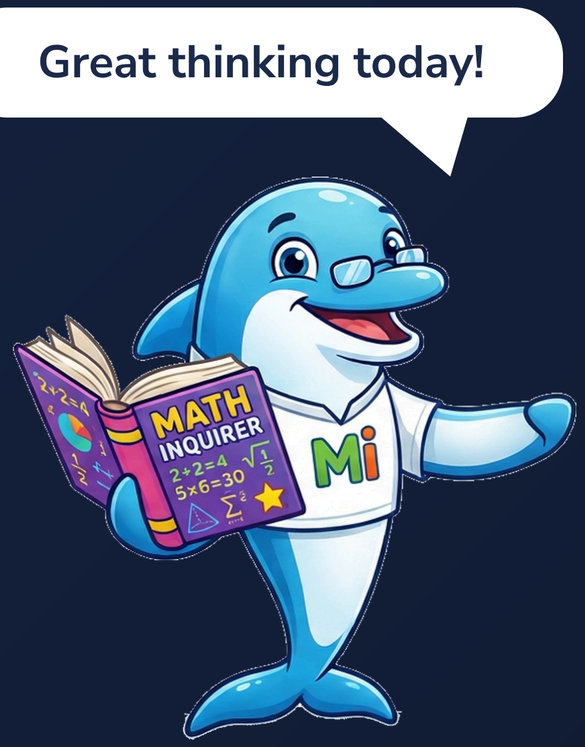
Well done, inquirers!

Show a grown-up one shape you folded today.

NEXT LESSON

Build the Figure

Have ready: the shape pieces page, cut out by a grown-up (20 pieces), a pencil and paper.



OPTIONAL

Optional: corner to corner

Mei folds a paper square corner to corner.

Each part is a half.

What shape is each part?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	Is a quarter always smaller than a half?
Make a puzzle	Fold a secret fold. Ask: half or quarter?
Teach it back	Show a toy how to check for halves.

Optional. Everyone may join.